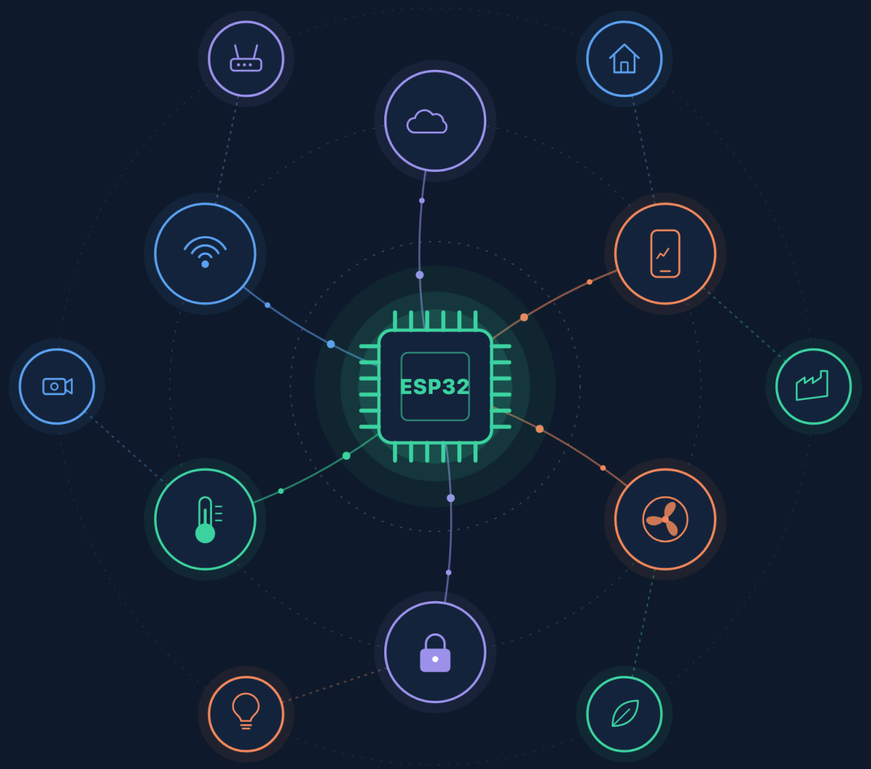
MODULE 4 · 45 MINUTES

Getting Online: Connectivity and Communication Protocols

How a tower site sends its data to the NOC: the road, the language and what it is worth. Then Lab 3.



Mehdi Lotfinejad

Introduction to the Internet of Things

THE ROAD AHEAD

Eight modules, one Smart Site Monitor

PART 1 — DEVICES AND DATA

01
What is IoT?
Lecture

02
How it's built
Lecture

03
Sensors and actuators
Lab 2

04
Getting online
You are here

PART 2 — CLOUD, AUTOMATION AND SECURITY

05
Cloud and dashboards
Lab 4

06
Automation
Lab 5

07
Security
Group activity

08
Real world + pitch
Lab 7

BY THE END OF THIS MODULE

You will be able to...

01 **Separate** the road (connection) from the language (protocol)

02 **Choose** a connection for a job, by reach and monthly cost

03 **Explain** what connectivity is worth to a telco

04 **Explain** MQTT as a group chat, with topics

05 **Put** your Smart Site Monitor online in Lab 3

THE BIG PICTURE

Two separate decisions

Protocol

The **language** of the messages: MQTT or HTTP

Network

Addresses and routing: the internet

Connection

The **road**: mobile network, Wi-Fi, cable, Bluetooth

Lab 3: **Wi-Fi** is the road, **MQTT** the language. A tower site: the **mobile network** is the road.

Part A

Choosing the road

Which connection for which job, what it is used for, and what it is worth.

THE ROADS, IN PLAIN WORDS

Which connection for which job

Connection	Good for	A telco example	Running cost
Mobile-network IoT (NB-IoT, LTE-M)	Small readings from far away, on a small battery	Fuel, battery and door readings at a tower site	A monthly plan per site
Satellite	Sites with no mobile coverage	A very remote tower site	A higher monthly plan
Cable	Reliable, where cables exist	Data-centre meters and cooling	Free once installed
Wi-Fi	Lots of data, indoors	Cameras in an exchange or office	Free on our own network
Bluetooth	Tiny batteries, near a phone	A shelter lock opened by phone	Free; uses a phone

Ask: **how far**, **how much power**, **how much data**, and **what does it cost each month?**



Photo: pslim, CC BY 2.0, via Wikimedia Commons

REMOTE SITE MONITORING

A tower site, watched from the NOC

- **Sense:** fuel, battery, door, shelter heat
- **Road:** the mobile network, with a SIM
- **Language:** a light one, such as MQTT
- **Check:** fuel drops while the generator is off?
- **Cost:** one SIM plan, or one site visit?

TELCO USE CASES

Three jobs for mobile-network IoT



Remote sites

Fuel, battery, door and shelter heat, seen from the NOC



Smart towers

Thousands of towers, their power and alarms, on one NOC screen



Asset tracking

Trackers on generators, batteries, spares and vans

In our data centres and exchanges, **cable** and **Wi-Fi** do this job indoors, with no monthly plan.

THE BUSINESS LENS

What connectivity is worth

A SIM, not a trip

Compare a monthly SIM plan with one site visit, hours away

Coverage

The mobile network already reaches remote sites. No cable to lay.

Scale

One network and one NOC screen for thousands of sites

The cost is real, but **small and predictable**: per device, per month.

THE BUSINESS LENS

Telcos sell IoT connectivity too

1 billion

NB-IoT and LTE-M connections worldwide, end of 2025

3

Maxis IoT plans in Malaysia: NB-IoT, 4G M2M, 5G IoT

Each connected device can mean **a SIM and a monthly plan**: a cost for the buyer, revenue for the telco.

Source: GSMA (connections, end of 2025); Maxis Business website (plans).

PAUSE AND CHECK

Pick a road: (a) a fuel sensor at a remote tower site; (b) a tracker on a generator moved between sites; (c) a small battery sensor read by a nearby phone.

Ask: How far? Mains or battery? How much data? What is already there? What does it cost per month?

Part B

The language: MQTT

How one reading from a site reaches everyone who needs it.

MQTT · THE LANGUAGE OF LAB 3

MQTT works like a group chat



The site posts

The site sends one message: “shelter at 29 °C”



The broker

Like the chat group: it passes every message on



Everyone gets it

The NOC, the alarm system and the on-call phone

Lab 3: **Wi-Fi** is the road, **MQTT** the language. At a tower site, only the road changes.

MQTT TOPICS

Topics and wildcards

Which messages does a subscription receive?

Topics are levels separated by "/". + matches exactly one level. # matches everything below.

Message published to topic…	subscribed to: iot-workshop/+/temperature	subscribed to: iot-workshop/#	subscribed to: iot-workshop/aisha/#
iot-workshop / aisha / temperature	✓	✓	✓
iot-workshop / aisha / humidity	✗	✓	✓
iot-workshop / ravi / temperature	✓	✓	✗
iot-workshop / ravi / humidity	✗	✓	✗
iot-workshop / ravi / room1 / temperature	✗	✓	✗

two levels where + stands → no

IN TELCO OPERATIONS

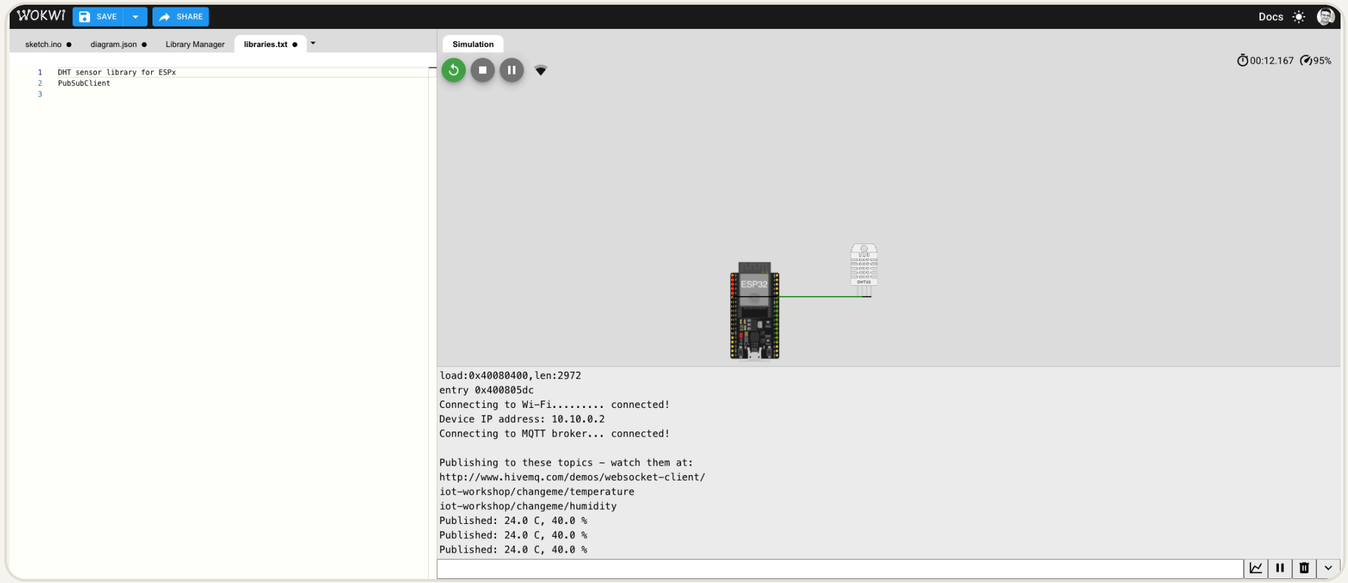
Getting sites online, in telco operations

Today	With IoT	The benefit
Site checks: drive out to look	Fuel, battery and door readings arrive by SIM	Fewer site visits
NOC alarms: passed on by phone	One message reaches the NOC, the alarm system and the on-call phone	Faster fault repair
Generators, spares: “where is it?”	Trackers show where it is, and if it moves	Less theft, less searching

Your team’s own numbers go on the canvas: costs in **box 6**, payback in **box 7** (Module 8).

LAB 3 · 23 MINUTES HANDS-ON

The tower site reports to the NOC



1. New builder: new ESP32 project in Wokwi
2. Paste sketch.ino — change “changeme” to your team name
3. Paste diagram.json and add libraries.txt
4. Press Play: Wi-Fi, then MQTT, then readings
5. Watch in HiveMQ, then fill in canvas box 3

Scan for the step-by-step guide
github.com/datatweets/intro-to-iot-workshop

KEY TAKEAWAYS

What to remember

- Two decisions: **the road** (connection) and **the language** (protocol)
- Pick the road by the job: **mobile-network IoT** at remote sites
- Connectivity pays when **a SIM plan replaces a site visit**
- Telcos **sell** IoT connectivity: 1 billion NB-IoT and LTE-M links
- **MQTT** is a group chat: the site posts once, everyone subscribed gets it

CHECK YOUR UNDERSTANDING

Quick quiz

Q1 A remote tower site has no Wi-Fi. How can its fuel sensor send data?

Q2 What is the difference between the road and the language?

Q3 A site publishes one MQTT message. Who receives it?

Q4 Does towers/+/temp match towers/north/site7/temp?

UP NEXT · MODULE 5

IoT Cloud Platforms and Data Visualisation

We give the site’s data a home: a live cloud chart in Lab 4, using HTTP this time.

Mehdi Lotfinejad

Book, labs and slides: github.com/datatweets/intro-to-iot-workshop